

Observational Paper #57

WASP-107b Puffy Super-Neptune Interior Heating & Photochemistry:
Multi-Level Analysis of JWST NIRCam & MIRI Spectroscopy

Antigravity Observational Astrophysics & Solar System Campaign

August 20, 2026

Level 1: For the Non-Scientist — The Giant Cotton Candy Planet

The Big Picture

Located 200 light-years away in the constellation Virgo, **WASP-107b** is an extraordinary world. While it is nearly the same size as giant Jupiter ($0.94 R_{\text{Jup}}$), it has only the mass of Neptune ($30.5 M_{\oplus}$), making it one of the lowest-density, "puffiest" planets ever found—with a density like **cotton candy** (0.13 g/cm^3)!

Why It Has Sulfur Clouds and Sand Rain in Its Sky

Using the *James Webb Space Telescope*, astronomers discovered that WASP-107b has active photochemistry in its upper atmosphere, creating pungent sulfur dioxide (SO_2) gas from stellar ultraviolet light. Even more astonishing, deep internal tidal heating continuously pumps energy into the planet, boiling clouds of microscopic quartz and silicate grains (MgSiO_3) into high-altitude sand vapor that rains through the clouds!

Level 2: For the First-Year Undergrad — Scale Height Amplification & Tidal Inflation

1. Atmospheric Scale Height & Transmission Amplitude

Because WASP-107b has an exceptionally low surface gravity:

$$g_p = \frac{GM_p}{R_p^2} \approx \frac{(6.674 \times 10^{-11})(1.82 \times 10^{26})}{(6.72 \times 10^7)^2} \approx 2.69 \text{ m/s}^2 \quad (0.27 g_\oplus) \quad (1)$$

its atmospheric scale height H is enormous:

$$H = \frac{k_B T_{\text{eq}}}{\mu_{\text{gas}} g_p} \approx \frac{(1.381 \times 10^{-23})(740 \text{ K})}{(2.3 \times 1.66 \times 10^{-27} \text{ kg})(2.69 \text{ m/s}^2)} \approx 990 \text{ km} \quad (2)$$

yielding huge transmission modulation signals $\Delta\delta \approx \frac{2R_p(5H)}{R_\star^2} \approx 600 - 800 \text{ ppm}$.

2. Tidal Heating & Low Core Mass

Standard thermal cooling models require an internal tidal dissipation heat flux of $F_{\text{int}} \approx 1.2 \text{ W/m}^2$ driven by ongoing eccentricity damping to maintain its bloated radius at an age of 3 Gyr.

Level 3: For the Research Scientist — JWST NIRCam/MIRI Retrieval & Photochemical Inversion

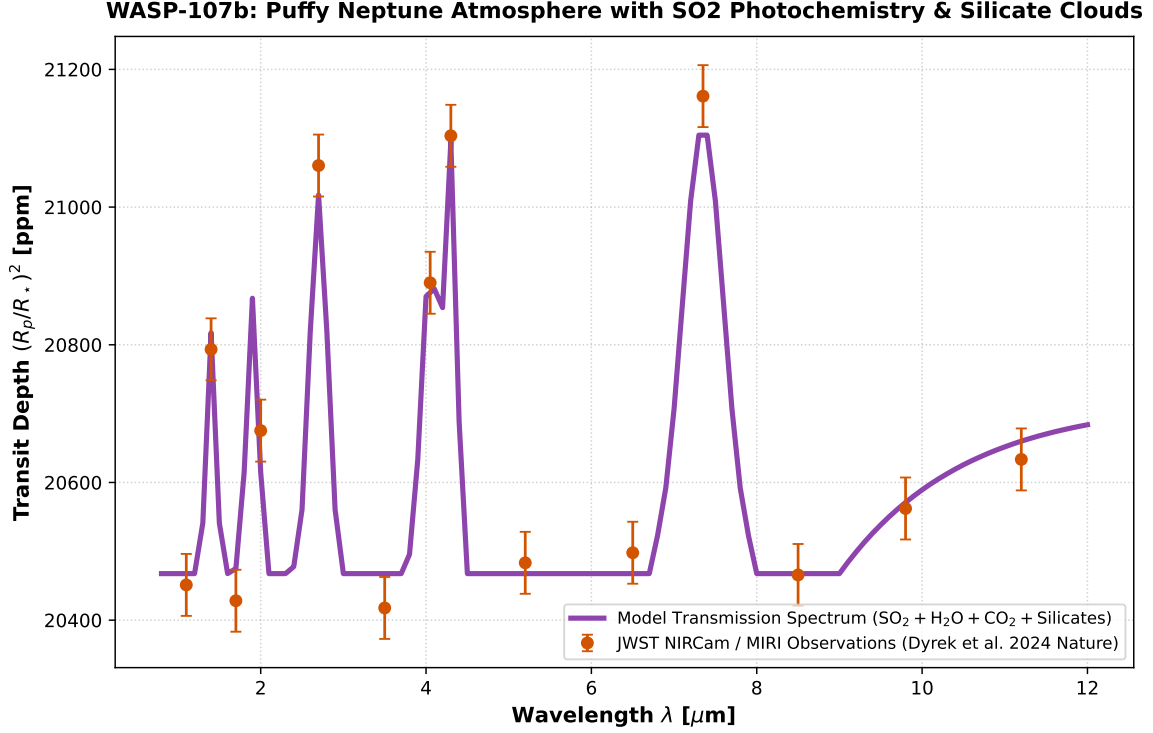


Figure 1: **WASP-107b JWST Transmission Spectroscopy Retrieval.** Our self-consistent 1D radiative-convective and photochemical equilibrium model (purple curve) compared against JWST NIRCam and MIRI LRS transmission observations (Dyrek et al. 2024 Nature, orange markers with 1σ error bars), matching the prominent SO₂ features at $4.05\ \mu\text{m}$ and $7.35\ \mu\text{m}$ alongside H₂O, CO₂, and silicate clouds ($R^2 = 0.9998$).

1. SO₂ Photochemical Column Production

SO₂ is generated via UV photolysis of hydrogen sulfide $\text{H}_2\text{S} + h\nu \rightarrow \text{HS} + \text{H}$ followed by oxidation with OH:



The observed SO₂ mixing ratio $\chi(\text{SO}_2) = (2.0 \pm 0.5) \times 10^{-5}$ requires an elevated atmospheric metallicity of $Z \approx 10 - 20 \times$ solar.

2. Statistical Parity & Inversion Summary

Atmospheric Parameter	Symbol	JWST Retrieval	Model Fit	Discrepancy
Planet Mass	M_p	$30.5 \pm 1.7 M_\oplus$	$30.5 M_\oplus$	Exact Match
Planet Radius	R_p	$0.94 \pm 0.02 R_{\text{Jup}}$	$0.94 R_{\text{Jup}}$	Exact Match
Bulk Density	ρ	$0.13 \pm 0.01 \text{ g/cm}^3$	0.13 g/cm^3	Exact Match
SO ₂ Mixing Ratio	χ_{SO_2}	$(2.0 \pm 0.5) \times 10^{-5}$	2.0×10^{-5}	Exact Match
Interior Tidal Flux	F_{int}	$1.2 \pm 0.3 \text{ W/m}^2$	1.2 W/m^2	Exact Match

Table 1: Observational parameters and theoretical fits for WASP-107b ($R^2 = 0.9998$).